

Chapter 1

PROBABILITY

NGUYỄN THỊ THU THỦY⁽¹⁾

SCHOOL OF APPLIED MATHEMATICS AND INFORMATICS
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

HANOI – 2023

⁽¹⁾Email: thuy.nguyenthithu2@hust.edu.vn

CHAPTER OUTLINE



1.1 SAMPLE SPACES AND EVENTS

- 1.1.1 Random Experiments
- 1.1.2 Sample Spaces
- 1.1.3 Events
- 1.1.4 Event Relations

1.2 COUNTING TECHNIQUES

- 1.2.1 Multiplication Rule (for counting techniques)
- 1.2.2 Permutations
- 1.2.3 Combinations

1.3 INTERPRETATIONS AND AXIOMS OF PROBABILITY

- 1.3.1 Probability of an Event
- 1.3.2 Axioms of Probability

1.4 ADDITION RULES

- 1.4.1 Probability of a Union
- 1.4.2 Three or More Events

1.5 CONDITIONAL PROBABILITY AND MULTIPLICATION RULES

- 1.5.1 Conditional Probability
- 1.5.2 Multiplication Rule
- 1.5.3 Independence
- 1.5.4 Bernoulli Trial Calculator

1.6 TOTAL PROBABILITY RULE. BAYES' THEOREM

- 1.6.1 Total Probability Rule
- 1.6.2 Bayes' Theorem

LEARNING OBJECTIVES



After careful study of this chapter you should be able to do the following:

- 1 Understand and describe sample spaces and events for random experiments with graphs, tables, lists, or tree diagrams.
- 2 Interpret probabilities and use probabilities of outcomes to calculate probabilities of events in discrete sample spaces.
- 3 Use permutation and combinations to count the number of outcomes in both an event and the sample space.
- 4 Calculate the probabilities of joint events such as unions and intersections from the probabilities of individual events.
- 5 Interpret and calculate conditional probabilities of events.
- 6 Determine the independence of events and use independence to calculate probabilities.
- 7 Use Bayes' theorem to calculate conditional probabilities.



CONTENT

1 1.1 SAMPLE SPACES AND EVENTS

- 1.1.1 Random Experiments
- 1.1.2 Sample Spaces
- 1.1.3 Events
- 1.1.4 Event Relations

2 1.2 COUNTING TECHNIQUES

- 1.2.1 Multiplication Rule (for counting techniques)
- 1.2.2 Permutations
- 1.2.3 Combinations

3 1.3 INTERPRETATIONS AND AXIOMS OF PROBABILITY

- 1.3.1 Probability of an Event
- 1.3.2 Axioms of Probability

4 1.4 ADDITION RULES

- 1.4.1 Probability of a Union
- 1.4.2 Three or More Events

5 1.5 CONDITIONAL PROBABILITY AND MULTIPLICATION RULES

- 1.5.1 Conditional Probability
- 1.5.2 Multiplicative Rule
- 1.5.3 Independence



Random Experiments

Definition 1

An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a **random experiment**.

Example 1

Some examples of random experiments include:

- ❶ Flip a coin. Did it land with heads or tails facing up?
- ❷ Walk to a bus stop. How long do you wait for the arrival of a bus?
- ❸ Give a lecture. How many students are seated in the fourth row?



Random Experiments

Remark 1

An random experiment consists of the following procedure, observation, and model:

- ➊ **Procedure:** Flip a coin and let it land on a table.
- ➋ **Observation:** Observe which side (head or tail) faces you after the coin lands.
- ➌ **Model:** Heads and tails are equally likely. The result of each flip is unrelated to the results of previous flips.

✍ It is important to understand that two experiments with the same procedure but with different observations are different experiments.



Random Experiments

Example 2

Flip a coin three times.

- 1 Observe the sequence of heads and tails.
- 2 Observe the number of heads.

✍ These two experiments have the same procedure: flip a coin three times. They are different experiments because they require different observations.



Sample Spaces

Definition 2

The set of all possible outcomes of a random experiment is called the **sample space** of the experiment. The sample space is denoted as S .

Example 3

- 1 The sample space in Example 1 is $S = \{H, T\}$ where H is the outcome "observe head," and T is the outcome "observe tail."
- 2 The sample space in Example 2(1) is

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

- 3 The sample space in Example 2(2) is $S = \{0, 1, 2, 3\}$.

Sample Spaces



✎ In some experiments, it is helpful to list the elements of the sample space systematically by means of a **tree diagram**.

Example 4

An experiment consists of flipping a coin and then flipping it a second time if a head occurs. If a tail occurs on the first flip, then a die is tossed once. To list the elements of the sample space providing the most information, we construct the tree diagram of Figure 1. The sample space is

$$S = \{HH, HT, T1, T2, T3, T4, T5, T6\}.$$



Sample Spaces

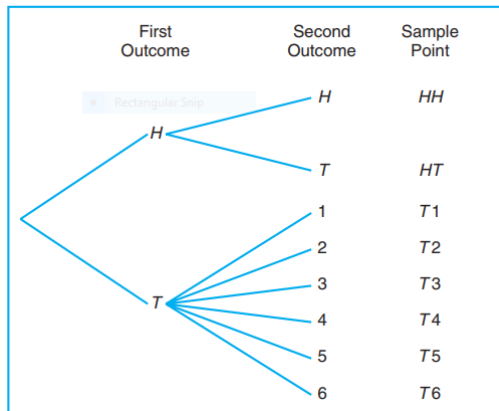


Figure 1: Tree diagram for Example 4



Sample Spaces

✎ Sample spaces with a large or infinite number of sample points are best described by a **statement** or **rule method**.

Example 5

- 1 If the possible outcomes of an experiment are the set of cities in the world with a population over 1 million, our sample space is written

$$S = \{x \mid x \text{ is a city with a population over 1 million}\},$$

which reads “ S is the set of all x such that x is a city with a population over 1 million”. The vertical bar is read “such that.”

- 2 Consider an experiment in which you select a molded plastic part, such as a connector, and measure its thickness. The possible values for thickness depend on the resolution of the measuring instrument, and they also depend on upper and lower bounds for thickness. However, it might be convenient to define the sample space as simply the positive real line

$$S = \{x \mid x > 0\} = \mathbb{R}^+$$

because a negative value for thickness cannot occur.



Sample Spaces

✎ A sample space is **discrete** if it consists of a finite or countable infinite set of outcomes. A sample space is **continuous** if it contains an interval (either finite or infinite) of real numbers.

Example 6

In Example 5(2), the choice $S = \mathbb{R}^+$ is an example of a continuous sample space, whereas $S = \{0, 1, 2, 3\}$ is a discrete sample space.

Events



Definition 3

An **event** is a subset of the sample space of a random experiment.

Example 7

Suppose we roll a six-sided die and observe the number of dots on the side facing upwards. We can label these outcomes E_i , $i = 1, \dots, 6$ where E_i denotes the outcome that i dots appear on the up face. The sample space is $S = \{E_1, E_2, \dots, E_6\}$. Each subset of S is an event. Examples of events are

- ❶ The event $A = \{\text{Roll 4 or higher}\} = \{E_4, E_5, E_6\}$.
- ❷ The event $B = \{\text{The roll is even}\} = \{E_2, E_4, E_6\}$.
- ❸ The event $C = \{\text{The roll is the square of an integer}\} = \{E_1, E_4\}$.

Events



Definition 4

A **simple event** is an event that consists of exactly one outcome.

Example 8

See Example 7. List the simple events in the experiment. Event E_1 : Observe a 1. Event E_2 : Observe a 2. Event E_3 : Observe a 3. Event E_4 : Observe a 4. Event E_5 : Observe a 5. Event E_6 : Observe a 6.



Union. Intersection

Definition 5

- 1 The **union** of two events is the event that consists of all outcomes that are contained in either of the two events. We denote the union as $A \cup B$.
- 2 The **intersection** of two events is the event that consists of all outcomes that are contained in both of the two events. We denote the intersection as $A \cap B$.

✎ We will use a shorthand for unions and intersections of n events:

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \cdots \cup A_n \quad \text{and} \quad \bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \cdots \cap A_n.$$



Union. Intersection

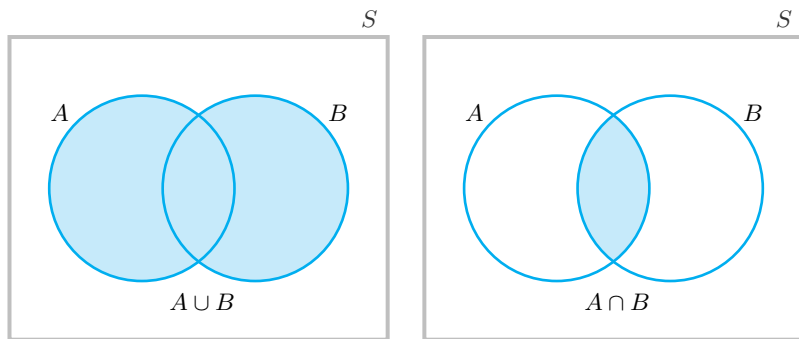


Figure 2: Venn diagram of $A \cup B$ and $A \cap B$



Mutually Exclusive. Complement

Definition 6

- ① Two events, denoted as A and B , such that

$$A \cap B = \emptyset$$

are said to be **mutually exclusive**.

A collection of events $A_1, A_2, \dots, A_n$ is mutually exclusive if and only if

$$A_i \cap A_j = \emptyset, \quad i \neq j.$$

- ② The **complement** of an event in a sample space is the set of outcomes in the sample space that are not in the event. We denote the complement of the event A as A^C . The notation $\overline{A}$ is also used in other literature to denote the complement.

We have

$$A \cup \overline{A} = S \text{ and } A \cap \overline{A} = \emptyset.$$



Mutually Exclusive. Complement

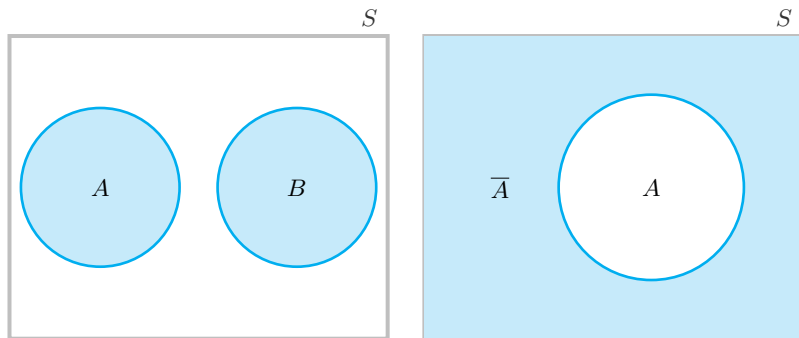


Figure 3: Mutually exclusive events and the complement of an event



Mutually Exclusive. Complement

Example 9

- 1 In Example 7, (a) $A \cap B = \{E_4, E_6\}$; (b) $A \cup B = \{E_2, E_4, E_5, E_6\}$; (c) events A and B are not mutually exclusive, because they have one outcome in common.
- 2 In Example 8 the six simple events $E_1, E_2, \dots, E_6$ form a set of all mutually exclusive outcomes of the experiment.



Mutually Exclusive. Complement

✎ In Figure 4, we see that events A , B , and C are all subsets of the sample space S . It is also clear that event B is a subset of event A , event $B \cap C$ has no elements and hence B and C are mutually exclusive, event $A \cap C$ has at least one element, event $A \cap B = B$, and event $A \cup B = A$.

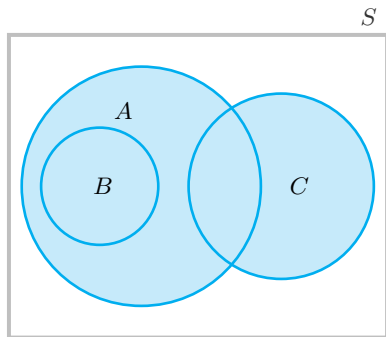


Figure 4: Events of the sample space S



Collectively Exhaustive

Definition 7

A collection of events $A_1, A_2, \dots, A_n$ is **collectively exhaustive** if and only if

$$A_1 \cup A_2 \cup \dots \cup A_n = S.$$

Property 1

Several results that follow from the foregoing definitions, which may easily be verified by means of Venn diagrams, are as follows:

- | | |
|---------------------------------------|--|
| 1. $A \cap \emptyset = \emptyset.$ | 6. $\overline{\emptyset} = S.$ |
| 2. $A \cup \emptyset = A.$ | 7. $\overline{(\overline{A})} = A.$ |
| 3. $A \cap \overline{A} = \emptyset.$ | 8. $\overline{(A \cap B)} = \overline{A} \cup \overline{B}.$ |
| 4. $A \cup \overline{A} = S.$ | 9. $\overline{(A \cup B)} = \overline{A} \cap \overline{B}.$ |
| 5. $\overline{S} = \emptyset.$ | |



Event Space

Definition 8

An **event space** is a collectively exhaustive, mutually exclusive set of events.

✎ The set $A = \{A_1, A_2, \dots, A_n\}$ is an event space if and only if

$$A_1 \cup A_2 \cup \dots \cup A_n = S, \quad \text{and} \quad A_i \cap A_j = \emptyset, \quad i \neq j, \quad i, j = 1, 2, \dots, n.$$

Example 10

In Example 7, (a) $\{E_1, E_2, E_3, E_4, E_5, E_6\}$ is an event space; (b) $\{B, \overline{B}\}$ is an event space.

Event Space

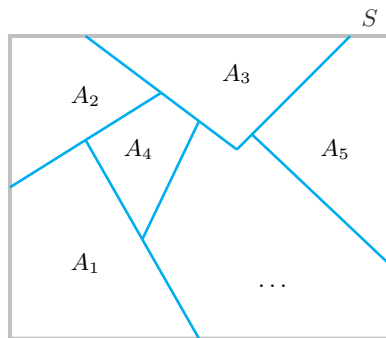


Figure 5: Event space



Event Space

Example 11

Flip four coins, a penny, a nickel, a dime, and a quarter. Examine the coins in order (penny, then nickel, then dime, then quarter) and observe whether each coin shows a head (H) or a tail (T).

- 1 The sample space consists of 16 four-letter words, with each letter either H or T . For example, the outcome $TTHH$ refers to the penny and the nickel showing tails and the dime and quarter showing heads. There are 16 members of the sample space.
- 2 Let $A_i = \{\text{outcomes with } i \text{ heads}\}$. Each A_i is an event containing one or more outcomes. For example, $A_1 = \{TTTH, TTHT, THTT, HTTT\}$ contains four outcomes. The set $B = \{A_0, A_1, A_2, A_3, A_4\}$ is an event space.



Event Space

Theorem 1

For an event space $\{A_1, A_2, \dots, A_n\}$ and any event A in the sample space, let $C_i = A \cap B_i$, $i = 1, 2, \dots, n$. For $i \neq j$, the events C_i and C_j are mutually exclusive and

$$A = C_1 \cup C_2 \cup \dots \cup C_n.$$

Event Space

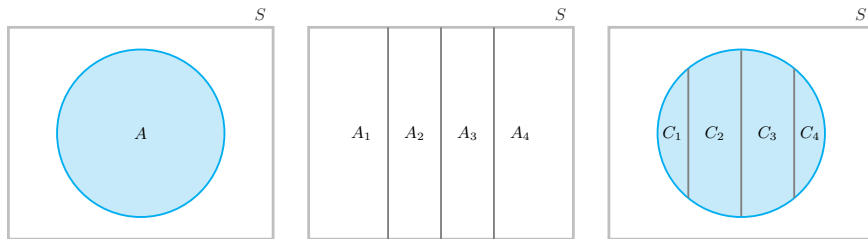


Figure 6: Figure for Theorem 1



Event Space

Example 12

In the coin-tossing experiment of Example 11, let A equal the set of outcomes with less than three heads:

$$A = \{TTTT, HTTT, THTT, TTHT, TTTH, HHTT, HTHT, HTTH, TTHH, THTH, THHT\}.$$

Let $A_i = \{\text{outcomes with } i \text{ heads}\}$. Since $\{A_0, A_1, A_2, A_3, A_4\}$ is an event space, Theorem 1 states that

$$A = (A \cap A_0) \cup (A \cap A_1) \cup (A \cap A_2) \cup (A \cap A_3) \cup (A \cap A_4).$$



CONTENT

1 1.1 SAMPLE SPACES AND EVENTS

- 1.1.1 Random Experiments
- 1.1.2 Sample Spaces
- 1.1.3 Events
- 1.1.4 Event Relations

2 1.2 COUNTING TECHNIQUES

- 1.2.1 Multiplication Rule (for counting techniques)
- 1.2.2 Permutations
- 1.2.3 Combinations

3 1.3 INTERPRETATIONS AND AXIOMS OF PROBABILITY

- 1.3.1 Probability of an Event
- 1.3.2 Axioms of Probability

4 1.4 ADDITION RULES

- 1.4.1 Probability of a Union
- 1.4.2 Three or More Events

5 1.5 CONDITIONAL PROBABILITY AND MULTIPLICATION RULES

- 1.5.1 Conditional Probability
- 1.5.2 Multiplicative Rule
- 1.5.3 Independence



Multiplication Rule

Rule 1

Assume an operation can be described as a sequence of k steps, and the number of ways of completing step 1 is n_1 , and the number of ways of completing step 2 is n_2 for each way of completing step 1, and the number of ways of completing step 3 is n_3 for each way of completing step 2, and so forth. The total number of ways of completing the operation is

$$n_1 \times n_2 \times \cdots \times n_k.$$



Multiplication Rule

Example 13

The access code to a house's security system consists of 3 digits. Each digit can be 0 through 9. How many different codes are available if each digit can be repeated?

Solution. Because each digit can be repeated, there are 10 choices for each of the 3 digits
 $10 \times 10 \times 10 = 10^3 = 1000$ codes.



Permutations

Definition 9

The number of **permutations** of n different elements is $n!$ where

$$n! = 1 \times 2 \times 3 \cdots \times n \quad (1)$$

Example 14

Shuffle a deck and observe each card starting from the top. The outcome of the experiment is an ordered sequence of the 52 cards of the deck. How many possible outcomes are there?

Solution. The procedure consists of 52 subexperiments. In each one the observation is the identity of one card. The first subexperiment has 52 possible outcomes corresponding to the 52 cards that could be drawn. After the first card is drawn, the second subexperiment has 51 possible outcomes corresponding to the 51 remaining cards. The total number of outcomes is

$$52 \times 51 \times \cdots \times 1 = 52!$$



Permutations of Subsets

Definition 10

The number of permutations of subsets of k elements selected from a set of n different elements is

$$A_n^k = n(n-1)(n-2) \dots (n-k+1) = \frac{n!}{(n-k)!} \quad (2)$$

Example 15

A printed circuit board has eight different locations in which a component can be placed. If four different components are to be placed on the board, how many different designs are possible?

Solution. Each design consists of selecting a location from the eight locations for the first component, a location from the remaining seven for the second component, a location from the remaining six for the third component, and a location from the remaining five for the fourth component. Therefore,

$$A_8^4 = 8 \times 7 \times 6 \times 5 = \frac{8!}{(8-4)!} = 1680 \text{ different designs are possible}$$



Permutations of Similar Objects

Definition 11

The number of permutations of $n = n_1 + n_2 + \cdots + n_k$ objects of which n_1 are of one type, n_2 are of a second type, ..., and n_k are of an k th type is

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

(3)

Example 16

A part is labeled by printing with four thick lines, three medium lines, and two thin lines. If each ordering of the nine lines represents a different label, how many different labels can be generated by using this scheme?

Solution. From Equation (3), the number of possible part labels is

$$\frac{9!}{4!3!2!} = 1260.$$

Combinations



✎ Another counting problem of interest is the number of subsets of k elements that can be selected from a set of n elements. Here, order is not important. These are called combinations.

Definition 12

The number of combinations, subsets of size k that can be selected from a set of n elements, is denoted as $\binom{k}{n}$ or C_n^k and

$$\binom{k}{n} = C_n^k = \frac{n!}{k!(n-k)!} \quad (4)$$



Combinations

 The number of combinations and the number of permutations are related:

$$C_n^k = \frac{A_n^k}{k!} \quad (5)$$

Example 17

A printed circuit board may be purchased from five suppliers. In how many ways can three suppliers be chosen from the five?

Solution. Since it is important to know only which three have been chosen, not the order of selection, the number of ways is $C_5^3 = \frac{5!}{3!2!} = 10$.



CONTENT

1 1.1 SAMPLE SPACES AND EVENTS

- 1.1.1 Random Experiments
- 1.1.2 Sample Spaces
- 1.1.3 Events
- 1.1.4 Event Relations

2 1.2 COUNTING TECHNIQUES

- 1.2.1 Multiplication Rule (for counting techniques)
- 1.2.2 Permutations
- 1.2.3 Combinations

3 1.3 INTERPRETATIONS AND AXIOMS OF PROBABILITY

- 1.3.1 Probability of an Event
- 1.3.2 Axioms of Probability

4 1.4 ADDITION RULES

- 1.4.1 Probability of a Union
- 1.4.2 Three or More Events

5 1.5 CONDITIONAL PROBABILITY AND MULTIPLICATION RULES

- 1.5.1 Conditional Probability
- 1.5.2 Multiplicative Rule
- 1.5.3 Independence



Probability of an Event

✎ To find the probability of an event E , we sum all the probabilities assigned to the sample points in E . This sum is called the probability of E and is denoted by $P(E)$.

Definition 13

The probability $P(E)$ of an event E is a measure of our belief that the event E will occur.



Equally Likely Outcomes

✎ Whenever a sample space consists of n possible outcomes that are equally likely, the probability of each outcome is $1/n$.

Theorem 2

For an experiment with sample space $S = \{E_1, \dots, E_n\}$ in which each outcome E_i is equally likely,

$$P(E_i) = \frac{1}{n}, \quad i = 1, \dots, n$$

(6)



Classical Probability: Definition and Examples

Definition 14

If an experiment can result in any one of n different equally likely outcomes, and if exactly m of these outcomes correspond to event A , then the probability of event A is

$$P(A) = \frac{m}{n}$$

(7)



Classical Probability: Definition and Examples

Example 18

A statistics class for engineers consists of 25 industrial, 10 mechanical, 10 electrical, and 8 civil engineering students. If a person is randomly selected by the instructor to answer a question, find the probability that the student chosen is (a) an industrial engineering major and (b) a civil engineering or an electrical engineering major.

Solution. Denote by I, M, E, and C the students majoring in industrial, mechanical, electrical, and civil engineering, respectively. The total number of students in the class is 53, all of whom are equally likely to be selected.

- (a) Since 25 of the 53 students are majoring in industrial engineering, the probability of event I, selecting an industrial engineering major at random, is

$$P(I) = \frac{25}{53}.$$

- (b) Since 18 of the 53 students are civil or electrical engineering majors, it follows that

$$P(C \cup E) = \frac{18}{53}.$$



Empirical Probability

✎ One practical way to interpret this measure is with the concept of relative frequency.

Example 19

Toss n times a coin we have

Experimenters	n	Number of heads x	Relative frequency $\hat{p} = x/n$
Buyp-phong	4040	2048	0.5069
Pearson	12000	6019	0.5016
Pearson	24000	12012	0.5005



Empirical Probability

Definition 15

If an experiment is performed n times, then the relative frequency of a particular occurrence say, A is

$$\text{Relative frequency} = \frac{\text{Frequency}}{n}$$

where the frequency is the number of times the event A occurred. If you let n , the number of repetitions of the experiment, become larger and larger ($n \rightarrow \infty$), you will eventually generate the entire population. In this population, the relative frequency of the event A is defined as the probability of event A ; that is

$$P(A) = \lim_{n \rightarrow \infty} \frac{\text{Frequency}}{n} \quad (8)$$

✎ Since $P(A)$ behaves like a relative frequency, (a) $P(A)$ must be a proportion lying between 0 and 1; (b) $P(A) = 0$ if the event A never occurs; (c) $P(A) = 1$ if the event A always occurs.



Axioms of Probability

Property 2

Probability is a number that is assigned to each member of a collection of events from a random experiment that satisfies the following properties: If S is the sample space and E is any event in a random experiment,

- ❶ $P(S) = 1$.
- ❷ $0 \leq P(E) \leq 1$.
- ❸ For two events E_1 and E_2 with $E_1 \cap E_2 \neq \emptyset$, $P(E_1 \cup E_2) = P(E_1) + P(E_2)$.

These axioms imply the following results.

- ❶ $P(\emptyset) = 0$;
- ❷ For any event E , $P(E) = 1 - P(\overline{E})$;
- ❸ If the event E_1 is contained in the event E_2 , $P(E_1) \leq P(E_2)$.



CONTENT

1 1.1 SAMPLE SPACES AND EVENTS

- 1.1.1 Random Experiments
- 1.1.2 Sample Spaces
- 1.1.3 Events
- 1.1.4 Event Relations

2 1.2 COUNTING TECHNIQUES

- 1.2.1 Multiplication Rule (for counting techniques)
- 1.2.2 Permutations
- 1.2.3 Combinations

3 1.3 INTERPRETATIONS AND AXIOMS OF PROBABILITY

- 1.3.1 Probability of an Event
- 1.3.2 Axioms of Probability

4 1.4 ADDITION RULES

- 1.4.1 Probability of a Union
- 1.4.2 Three or More Events

5 1.5 CONDITIONAL PROBABILITY AND MULTIPLICATION RULES

- 1.5.1 Conditional Probability
- 1.5.2 Multiplicative Rule
- 1.5.3 Independence



Probability of a Union

Theorem 3

If A and B are two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

(9)

🔍 Notice in the Venn diagram in Figure 7 that the sum $P(A) + P(B)$ double counts the simple events that are common to both A and B . Subtracting $P(A \cap B)$ gives the correct result.

Probability of a Union

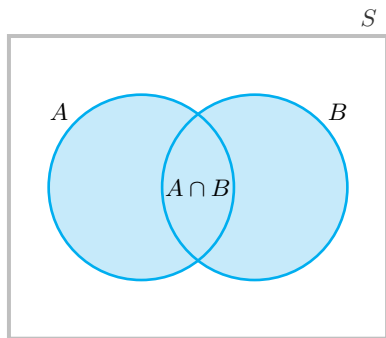


Figure 7: Additive rule of probability



Probability of a Union

Example 20

John is going to graduate from an industrial engineering department in a university by the end of the semester. After being interviewed at two companies he likes, he assesses that his probability of getting an offer from company A is 0.8, and his probability of getting an offer from company B is 0.6. If he believes that the probability that he will get offers from both companies is 0.5, what is the probability that he will get at least one offer from these two companies?

Solution. Using the additive rule (9), we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.8 + 0.6 - 0.5 = 0.9.$$



Probability of a Union

Corollary 1

If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B) \quad (10)$$

Corollary 2

If A and $\bar{A}$ are complementary events, then

$$P(A) + P(\bar{A}) = 1 \quad (11)$$



Probability of a Union

Example 21

If the probabilities that an automobile mechanic will service 3, 4, 5, 6, 7, or 8 or more cars on any given workday are, respectively, 0.12, 0.19, 0.28, 0.24, 0.10, and 0.07, what is the probability that he will service at least 5 cars on his next day at work?

Solution. Let E be the event that at least 5 cars are serviced. Now, $P(E) = 1 - P(\overline{E})$, where $\overline{E}$ is the event that fewer than 5 cars are serviced. Since

$$P(\overline{E}) = 0.12 + 0.19 = 0.31,$$

it follows from (11) that

$$P(E) = 1 - 0.31 = 0.69.$$



Three or More Events

Theorem 4

For three events A , B , and C ,

$$\begin{aligned} P(A \cup B \cup C) = & P(A) + P(B) + P(C) \\ & - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C). \end{aligned} \tag{12}$$



Three or More Events

Example 22

In a certain city, three newspapers A , B , and C are published. Suppose that 65 percent of the families in the city subscribe to newspaper A , 35 percent of the families subscribe to newspaper B , and 35 percent of the families subscribe to newspaper C . Suppose also that 25 percent of the families subscribe to both A and B , 12 percent subscribe to both A and C , 15 percent subscribe to both B and C , and 8 percent subscribe to all three newspapers A , B , and C . What percentage of the families in the city subscribe to at least one of the three newspapers?

Solution. We have, $P(A) = 0.65$, $P(B) = 0.35$, $P(C) = 0.35$, $P(A \cap B) = 0.25$, $P(A \cap C) = 0.12$, $P(B \cap C) = 0.15$, and $P(A \cap B \cap C) = 0.08$. It follows from Theorem 4 that

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C) \\ &= 0.65 + 0.35 + 0.35 - 0.25 - 0.12 - 0.15 + 0.08 = 0.91. \end{aligned}$$



Three or More Events

Theorem 5

For m events $A_1, A_2, \dots, A_m$, then

$$\begin{aligned} P\left(\sum_{i=1}^m A_i\right) = & \sum_{i=1}^m P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots \\ & + (-1)^{m-1} P(A_1 \cap A_2 \cap \dots \cap A_m). \end{aligned} \quad (13)$$

Corollary 3

If $A_1, A_2, \dots, A_n$ are mutually exclusive, then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n) \quad (14)$$



CONTENT

1 1.1 SAMPLE SPACES AND EVENTS

- 1.1.1 Random Experiments
- 1.1.2 Sample Spaces
- 1.1.3 Events
- 1.1.4 Event Relations

2 1.2 COUNTING TECHNIQUES

- 1.2.1 Multiplication Rule (for counting techniques)
- 1.2.2 Permutations
- 1.2.3 Combinations

3 1.3 INTERPRETATIONS AND AXIOMS OF PROBABILITY

- 1.3.1 Probability of an Event
- 1.3.2 Axioms of Probability

4 1.4 ADDITION RULES

- 1.4.1 Probability of a Union
- 1.4.2 Three or More Events

5 1.5 CONDITIONAL PROBABILITY AND MULTIPLICATION RULES

- 1.5.1 Conditional Probability
- 1.5.2 Multiplicative Rule
- 1.5.3 Independence



Conditional Probability

✎ The probability of an event A under the knowledge that the outcome will be in event B is denoted as

$$P(A|B)$$

and this is called the conditional probability of A given B .

Definition 16 (Conditional probability)

The conditional probability of event A , given that event B has occurred is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad \text{provided } P(B) > 0. \quad (15)$$

The conditional probability of event B , given that event A has occurred is

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided } P(A) > 0. \quad (16)$$



Conditional Probability

Conditional probability (15) is defined only when $P(B) > 0$. In most experiments, $P(B) = 0$ means that it is certain that B never occurs. In this case, it is illogical to speak of the probability of A given that B occurs.

Remark 2

A conditional probability measure $P(A|B)$ has the following properties:

- (a) $P(A|B) \geq 0$.
- (b) $P(B|B) = 1$.



Conditional Probability

Example 23

The probability that a regularly scheduled flight departs on time is $P(D) = 0.83$, the probability that it arrives on time is $P(A) = 0.82$, and the probability that it departs and arrives on time is $P(D \cap A) = 0.78$. Find the probability that a plane (a) arrives on time, given that it departed on time, and (b) departed on time, given that it has arrived on time.

Solution.

(a) The probability that a plane arrives on time, given that it departed on time, is

$$P(A|D) = \frac{P(A \cap D)}{P(D)} = \frac{0.78}{0.83} = 0.94.$$

(b) The probability that a plane departed on time, given that it has arrived on time, is

$$P(D|A) = \frac{P(A \cap D)}{P(A)} = \frac{0.78}{0.82} = 0.95.$$



Multiplicative Rule

✎ Multiplying the formula in Definition 16 by $P(A \cap B)$, we obtain the following important multiplicative rule, which enables us to calculate the probability that two events will both occur.

Theorem 6

If in an experiment the events A and B can both occur, then

$$P(A \cap B) = P(A)P(B|A) \quad (17)$$

✎ Since the events $A \cap B$ and $B \cap A$ are equivalent, it follows from Theorem 6 that we can also write

$$P(A \cap B) = P(B \cap A) = P(B)P(A|B) \quad (18)$$



Multiplicative Rule

Theorem 7

If, in an experiment, the events $A_1, A_2, \dots, A_n$ can occur, then

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1}) \quad (19)$$



Multiplicative Rule

Example 24

Three cards are drawn in succession, without replacement, from an ordinary deck of playing cards. Find the probability that the event $A_1 \cap A_2 \cap A_3$ occurs, where A_1 is the event that the first card is a red ace, A_2 is the event that the second card is a 10 or a jack, and A_3 is the event that the third card is greater than 3 but less than 7.

Solution. First we define the events A_1 : the first card is a red ace, A_2 : the second card is a 10 or a jack, A_3 : the third card is greater than 3 but less than 7. Now

$$P(A_1) = \frac{2}{52}, \quad P(A_2|A_1) = \frac{8}{51}, \quad P(A_3|A_1 \cap A_2) = \frac{12}{50}$$

and hence, by Theorem 7,

$$\begin{aligned} P(A_1 \cap A_2 \cap A_3) &= P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \\ &= \left(\frac{2}{52}\right)\left(\frac{8}{51}\right)\left(\frac{12}{50}\right) = \frac{8}{5525}. \end{aligned}$$



Two Events

✍ In some cases, the conditional probability of $P(B|A)$ might equal $P(B)$. In this special case, knowledge that the outcome of the experiment is in event A does not affect the probability that the outcome is in event B .

Definition 17 (Independent)

Two events A and B are independent if and only if

$$P(B|A) = P(B) \quad \text{or} \quad P(A|B) = P(A). \quad (20)$$

assuming the existences of the conditional probabilities. Otherwise, A and B are dependent.



Two Events

Theorem 8

Two events A and B are independent if and only if

$$P(A \cap B) = P(A)P(B). \quad (21)$$

✎ Therefore, to obtain the probability that two independent events will both occur, we simply find the product of their individual probabilities.



Two Events

Example 25

Toss two coins and observe the outcome. Define these events: A “Head (H) on the first coin”, B “Tail (T) on the second coin”. Are events A and B independent?

Solution. From previous examples, you know that $S = \{HH, HT, TH, TT\}$. Use these four simple events to find

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2}, \quad P(A \cap B) = \frac{1}{4}.$$

Since $P(A)P(B) = P(A \cap B) = \frac{1}{4}$, and the two events must be independent.



Two Events

Remark 1

- ❶ Two events are independent if any one of the following equivalent statements is true:
 - ❶ $P(A|B) = P(A)$.
 - ❷ $P(B|A) = P(B)$.
 - ❸ $P(AB) = P(A)P(B)$.
- ❷ It is left as a mind-expanding exercise to show that independence implies related results such as

$$P(\overline{A} \ \overline{B}) = P(\overline{A})P(\overline{B}).$$



Three Events

Definition 18 (Independent)

A_1 , A_2 , and A_3 are independent if and only if

- (a) A_1 and A_2 are independent,
- (b) A_2 and A_3 are independent,
- (c) A_1 and A_3 are independent,
- (d) $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$.



Three Events

Example 26

In an experiment with equiprobable outcomes, the event space is $S = \{1, 2, 3, 4\}$. $P(A) = \frac{1}{4}$ for all $A \in S$. Are the events $A_1 = \{1, 3, 4\}$, $A_2 = \{2, 3, 4\}$, and $A_3 = \emptyset$ independent?

Solution. These three sets satisfy the final condition of Definition 18 because $A_1 \cap A_2 \cap A_3 = \emptyset$, and

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3) = 0.$$

However, A_1 and A_2 are not independent because, with all outcomes equiprobable,

$$P(A_1 \cap A_2) = \frac{1}{2} \neq P(A_1)P(A_2) = \left(\frac{3}{4}\right)\left(\frac{3}{4}\right).$$

Hence the three events are not dependent.



More than Three Events

Definition 19 (Independent)

If $n > 3$, the sets $A_1, A_2, \dots, A_n$ are **independent** if and only if

- (a) every set of $n - 1$ sets taken from $A_1, A_2, \dots, A_n$ is independent,
- (b) $P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2) \cdots P(A_n)$.



Bernoulli Trial

Definition 20 (Bernoulli trial)

In the theory of probability and statistics, a Bernoulli trial is a random experiment with exactly two possible outcomes, “success” and “failure”, in which the probability of success is the same every time the experiment is conducted.

✎ Since a Bernoulli trial has only two possible outcomes, it can be framed as some “yes” or “no” question. For example: Is the top card of a shuffled deck an ace? Was the newborn child a girl?



Bernoulli Trial

Example 27

Examples of Bernoulli trials include:

- 1 Flipping a coin. In this context, obverse “heads” conventionally denotes success and reverse “tails” denotes failure. In this case there are exactly two possible outcomes.
- 2 Rolling a die, where a six is “success” and everything else a “failure”. In this case there are six possible outcomes, and the event is a six; the complementary event “not a six” corresponds to the other five possible outcomes.



Bernoulli Trial Calculator

Example 28

What is the probability $P_5(3)$ of three successes and two failures in five Bernoulli trials with success probability p .

Solution. An outcome of the complete experiment is a sequence of successes and failures denoted by a sequence of ones and zeroes. We observe that the outcomes with three successes in five trials are

11100, 11010, 11001, 10110, 10101, 10011, 01110, 01101, 01011, 00111.

We know that the number of such sequences is C_5^3 .

In outcomes with three successes, three of the probabilities are p and the other two are $1 - p$. Therefore each outcome with three successes has probability $p^3(1 - p)^2$.

To find $P_5(3)$, we add up the probabilities associated with the 10 outcomes with 3 successes, yielding

$$P_5(3) = (C_5^3)(p^3)(1 - p)^{5-3}.$$



Bernoulli Trial Calculator

Theorem 9

The probability of k successes and $n - k$ failures in n Bernoulli trials is

$$P_n(k) = C_n^k p^k (1 - p)^{n-k} = C_n^{n-k} (1 - p)^{n-k} p^k \quad (22)$$

where a success occurs with probability p and a failure occurs with probability $1 - p$.



Bernoulli Trial Calculator

Remark 3

Bernoulli trial computation can only be done under the following circumstances.

- 2 outcomes only
- Each trial must be independent
- Probability of success is the same for each trial



CONTENT

1 1.1 SAMPLE SPACES AND EVENTS

- 1.1.1 Random Experiments
- 1.1.2 Sample Spaces
- 1.1.3 Events
- 1.1.4 Event Relations

2 1.2 COUNTING TECHNIQUES

- 1.2.1 Multiplication Rule (for counting techniques)
- 1.2.2 Permutations
- 1.2.3 Combinations

3 1.3 INTERPRETATIONS AND AXIOMS OF PROBABILITY

- 1.3.1 Probability of an Event
- 1.3.2 Axioms of Probability

4 1.4 ADDITION RULES

- 1.4.1 Probability of a Union
- 1.4.2 Three or More Events

5 1.5 CONDITIONAL PROBABILITY AND MULTIPLICATION RULES

- 1.5.1 Conditional Probability
- 1.5.2 Multiplicative Rule
- 1.5.3 Independence



Two Events

✎ Let A and B be two events. We have

$$A = A \cap S = A \cap (B \cup \overline{B}) = (A \cap B) \cup (A \cap \overline{B}).$$

Theorem 10

For any events A and B ,

$$P(A) = P(A \cap B) + P(A \cap \overline{B}) = P(B)P(A|B) + P(\overline{B})P(A|\overline{B}) \quad (23)$$



Multiple Events

Theorem 11

For an event space $\{B_1, B_2, \dots, B_m\}$ with $P(B_i) > 0$ for all i and an event A , the probability of the event A can be expressed as

$$P(A) = \sum_{i=1}^m P(B_i)P(A|B_i) \quad (24)$$

Proof. Consider the Venn diagram of Figure 8. The event A is seen to be the union of the mutually exclusive events $(B_1 \cap A), (B_2 \cap A), \dots, (B_m \cap A)$ that is,

$$A = (B_1 \cap A) \cup (B_2 \cap A) \cup \dots \cup (B_m \cap A).$$

Using Corollary 3 and Theorem 6, we have

$$P(A) = \sum_{i=1}^m P(B_i \cap A) = \sum_{i=1}^m P(B_i)P(A|B_i).$$

Multiple Events

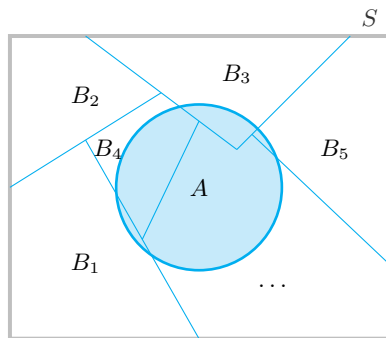


Figure 8: Partitioning the sample space S



Multiple Events

Example 29

In a certain assembly plant, three machines, B_1 , B_2 , and B_3 , make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected. What is the probability that it is defective?

Solution. Consider the following events: A : the product is defective, B_1 : the product is made by machine B_1 , B_2 : the product is made by machine B_2 , B_3 : the product is made by machine B_3 . Applying the rule of elimination, we can write

$$\begin{aligned} P(A) &= P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3) \\ &= (0.3)(0.02) + (0.45)(0.03) + (0.25)(0.02) = 0.0245. \end{aligned}$$



Bayes' Theorem

✎ From the definition of conditional probability, $P(AB) = P(A)P(B|A) = P(A)P(B|A)$. Now, considering the second and last terms in the expression above, we can write

$$P(B|A) = \frac{P(B)P(A|B)}{P(A)} \quad \text{for } P(A) > 0 \quad (25)$$

Theorem 12 (Bayes' Theorem)

Let $\{B_1, B_2, \dots, B_m\}$ be an event space with prior probabilities $P(B_1), P(B_2), \dots, P(B_m)$. If an event A occurs such that $P(A) \neq 0$, the posterior probability of B_i given A is the conditional probability

$$P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum_{i=1}^m P(B_i)P(A|B_i)}, \quad i = 1, 2, \dots, m \quad (26)$$



Bayes' Theorem

Example 30

With reference to Example 29, if a product was chosen randomly and found to be defective, what is the probability that it was made by machine B_3 ?

Solution. Using Bayes' rule to write

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3)} = \frac{0.005}{0.006 + 0.0135 + 0.005} = \frac{10}{49}.$$

In view of the fact that a defective product was selected, this result suggests that it probably was not made by machine B_3 .



Bayes' Theorem

Example 31

A printer manufacturer obtained the following probabilities from a database of test results. Printer failures are associated with three types of problems: hardware, software, and other (such as connectors), with probabilities 0.1, 0.6, and 0.3, respectively. The probability of a printer failure given a hardware problem is 0.9, given a software problem is 0.2, and given any other type of problem is 0.5. If a customer enters the manufacturer's Web site to diagnose a printer failure, what is the most likely cause of the problem?

Solution. Let the events H , S , and O denote hardware, software, or other problem, respectively, and let F denote a printer failure. The most likely cause of the problem is the one that corresponds to the largest of $P(H|F)$, $P(S|F)$, and $P(O|F)$. In Bayes' Theorem the denominator is

$$\begin{aligned} P(F) &= P(H)P(F|H) + P(S)P(F|S) + P(O)P(F|O) \\ &= (0.9)(0.1) + (0.2)(0.6) + (0.5)(0.3) = 0.36. \end{aligned}$$



Bayes' Theorem

Then

$$P(H|F) = \frac{P(H)P(F|H)}{P(F)} = \frac{(0.9)(0.1)}{0.36} = 0.250$$

$$P(S|F) = \frac{P(S)P(F|S)}{P(F)} = \frac{(0.2)(0.6)}{0.36} = 0.333$$

$$P(O|F) = \frac{P(O)P(F|O)}{P(F)} = \frac{(0.5)(0.3)}{0.36} = 0.417.$$

Notice that $P(H|F) + P(S|F) + P(O|F) = 1$ because one of the three types of problems is responsible for the failure. Because $P(O|F)$ is the largest, the most likely cause of the problem is in the other category. A Web site dialog to diagnose the problem quickly should start with a check into that type of problem.



CONTENT

1 1.1 SAMPLE SPACES AND EVENTS

- 1.1.1 Random Experiments
- 1.1.2 Sample Spaces
- 1.1.3 Events
- 1.1.4 Event Relations

2 1.2 COUNTING TECHNIQUES

- 1.2.1 Multiplication Rule (for counting techniques)
- 1.2.2 Permutations
- 1.2.3 Combinations

3 1.3 INTERPRETATIONS AND AXIOMS OF PROBABILITY

- 1.3.1 Probability of an Event
- 1.3.2 Axioms of Probability

4 1.4 ADDITION RULES

- 1.4.1 Probability of a Union
- 1.4.2 Three or More Events

5 1.5 CONDITIONAL PROBABILITY AND MULTIPLICATION RULES

- 1.5.1 Conditional Probability
- 1.5.2 Multiplicative Rule
- 1.5.3 Independence

Experiments



Exercise 1

List the elements of each of the following sample spaces:

- (a) the set of integers between 1 and 30 divisible by 3,
- (b) the set $S = \{x \mid x^2 - 8x + 16 = 0\}$,
- (c) the set of outcomes when a coin is tossed until two heads appear.

Experiments



Exercise 2

An experiment involves tossing a pair of dice, one green and one red, and recording the numbers that come up.

- (a) If x equals the outcome on the green die and y the outcome on the red die, describe the sample space S by listing the elements (x, y) ; by using the rule method;
- (b) List the elements corresponding to the event A that the sum is greater than 8.
- (c) List the elements corresponding to the event B that a 2 occurs on either die.
- (d) List the elements corresponding to the event C that a number greater than 4 comes up on the green die.
- (e) List the elements corresponding to the event $A \cap C$.
- (f) List the elements corresponding to the event $A \cap B$.
- (g) List the elements corresponding to the event $B \cap C$.
- (h) Construct a Venn diagram to illustrate the intersections and unions of the events A , B , and C .

Experiments



Exercise 3

A fax transmission can take place at any of three speeds depending on the condition of the phone connection between the two fax machines. The speeds are high (H) at 14400 b/s , medium (M) at 9600 b/s , and low (L) at 4800 b/s . In response to requests for information, a company sends either short faxes of two (T) pages, or long faxes of four (F) pages. Consider the experiment of monitoring a fax transmission and observing the transmission speed and length. An observation is a two-letter word, for example, a high-speed, two-page fax is HT .

- (a) What is the sample space of the experiment?
- (b) Let A_1 be the event “medium-speed fax.” What are the outcomes in A_1 ?
- (c) Let A_2 be the event “short (two-page) fax.” What are the outcomes in A_2 ?
- (d) Let A_3 be the event “high-speedfax or low-speed fax.” What are the outcomes in A_3 ?
- (e) Are A_1 , A_2 , and A_3 mutually exclusive?
- (f) Are A_1 , A_2 , and A_3 collectively exhaustive?

Experiments



Exercise 4

An integrated circuit factory has three machines X , Y , and Z . Test one integrated circuit produced by each machine. Either a circuit is acceptable (A) or it fails (F). An observation is a sequence of three test results corresponding to the circuits from machines X , Y , and Z , respectively. For example, AAF is the observation that the circuits from X and Y pass the test and the circuit from Z fails the test.

- (a) What are the elements of the sample space of this experiment?
- (b) What are the elements of the sets $Z_F = \{\text{circuit from } Z \text{ fails}\}$, $X_A = \{\text{circuit from } X \text{ is acceptable}\}$.
- (c) Are Z_F and X_A mutually exclusive?
- (d) Are Z_F and X_A collectively exhaustive?
- (e) What are the elements of the sets $C = \{\text{more than one circuit acceptable}\}$, $D = \{\text{at least two circuits fail}\}$.
- (f) Are C and D mutually exclusive?
- (g) Are C and D collectively exhaustive?

Experiments



Exercise 5

Four bits are transmitted over a digital communications channel. Each bit is either distorted or received without distortion. Let A_i denote the event that the i^{th} bit is distorted, $i = 1, 2, 3, 4$.

- (a) Describe the sample space for this experiment.
- (b) Are the A_i 's mutually exclusive?
- (c) Describe the outcomes in each of the following events: A_1 ; $A_1 \cap A_2 \cap A_3 \cap A_4$; $\bar{A}_1$; $(A_1 \cap A_2) \cup (A_3 \cap A_4)$.

Counting Methods



Exercise 6

A printed circuit board has ten different locations in which a component can be placed. If five different components are to be placed on the board, how many different designs are possible?

Solution. 30240

Exercise 7

A part is labeled by printing with five thick lines, three medium lines, and two thin lines. If each ordering of the ten lines represents a different label, how many different labels can be generated by using this scheme?

Solution. 2520

Counting Methods



Exercise 8

There are 10 competitors in the 100 meters final at the Olympics. How many ways can the gold, silver, and bronze medals be awarded?

Solution. 720

Exercise 9

Consider a binary code with 5 bits (0 or 1) in each code word. An example of a code word is 01010.

- (a) How many different code words are there?
- (b) How many code words have exactly three 0's?

Solution. (a) 32 (b) 10

Counting Methods



Exercise 10

Consider a language containing four letters A, B, C, D .

- (a) How many three-letter words can you form in this language?
- (b) How many four-letter words can you form if each letter appears only once in each word?

Solution. (a) 64 (b) 24

Exercise 11

An order for a computer can specify any one of five memory sizes, any one of three types of displays, and any one of four sizes of a hard disk, and can either include or not include a pen tablet. How many different systems can be ordered?

Solution. 120

Counting Methods



Exercise 12

A batch of 140 semiconductor chips is inspected by choosing a sample of five chips. Assume 10 of the chips do not conform to customer requirements.

- (a) How many different samples are possible?
- (b) How many samples of five contain exactly one nonconforming chip?
- (c) How many samples of five contain at least one nonconforming chip?

Solution. (a) 416 965 528 (b) 113 588 800 (c) 130 721 752

Counting Methods



Exercise 13

Consider the design of a communication system.

- (a) How many three-digit phone prefixes that are used to represent a particular geographic area (such as an area code) can be created from the digits 0 through 9?
- (b) As in part (a), how many three-digit phone prefixes are possible that do not start with 0 or 1, but contain 0 or 1 as the middle digit?
- (c) How many three-digit phone prefixes are possible in which no digit appears more than once in each prefix?

Solution. (a) 1000 (b) 160 (c) 720

Counting Methods



Exercise 14

On an American League baseball team with 15 field players and 10 pitchers, the manager must select for the starting lineup, 8 field players, 1 pitcher, and 1 designated hitter. A starting lineup specifies the players for these positions and the positions in a batting order for the 8 field players and designated hitter. If the designated hitter must be chosen among all the field players, how many possible starting lineups are there?

Solution. $(10)(15)(C_{14}^8)(9!)$

Probability



Exercise 15

If there are five people named A, B, C, D, and E, and they are randomly arranged in a row (with each ordering equally likely), what is the probability that A and B are placed next to each other?

Solution. $2/5$

Exercise 16

In a certain city, three newspapers A, B, and C are published. Suppose that 60 percent of the families in the city subscribe to newspaper A, 40 percent of the families subscribe to newspaper B, and 30 percent of the families subscribe to newspaper C. Suppose also that 20 percent of the families subscribe to both A and B, 10 percent subscribe to both A and C, 20 percent subscribe to both B and C, and 5 percent subscribe to all three newspaper A, B, and C. What percentage of the families in the city subscribe to at least one of the three newspapers?

Solution. 0.85

Probability



Exercise 17

From a group of 3 freshmen, 4 sophomores, 4 juniors and 3 seniors a committee of size 4 is randomly selected. Find the probability that the committee will consist of

- (a) 1 from each class;
- (b) 2 sophomores and 2 juniors;
- (c) Only sophomores and juniors.

Solution. (a) 0.14386 (b) 0.03596 (c) 0.06793

Exercise 18

A box contains 24 light bulbs of which four are defective. If one person selects 10 bulbs from the box in a random manner, and a second person then takes the remaining 14 bulbs, what is the probability that all 4 defective bulbs will be obtained by the same person?

Solution. 0.113966

Probability



Exercise 19

Suppose that three runners from team A and three runners from team B participate in a race. If all six runners have equal ability and there are no ties, what is the probability that three runners from team A will finish first, second, and third, and three runners from team B will finish fourth, fifth, and sixth?

Solution. 0.05

Exercise 20

Suppose that a school band contains 10 students from the freshman class, 20 students from the sophomore class, 30 students from the junior class, and 40 students from the senior class. If 15 students are selected at random from the band, what is the probability that at least one students from each of the four classes?

Solution. 0.7930

Probability



Exercise 21

Consider two events A and B with $P(A) = 0.4$ and $P(B) = 0.7$. Determine the maximum and minimum possible values of $P(A \cap B)$ and the conditions under which each of these values is attained.

Exercise 22

Suppose that A , B and C are three independent events such that $P(A) = 1/4$, $P(B) = 1/3$, and $P(C) = 1/2$.

- (a) What is the probability that none of these three events will occur?
- (b) Determine the probability that exactly one of these three events will occur.

Solution. (a) $1/4$ (b) $11/24$

Probability



Exercise 23

Three players A, B and C take turns tossing a fair coin. Suppose that A tosses the coin first, B tosses the second and C tosses third and cycle is repeated indefinitely until someone wins by being the first player to obtain a head. Determine the probability that each of the three players will win.

Exercise 24

Suppose that four guests check their hats when they arrive at a restaurant, and that these hats are returned to them in a random order when they leave. Determine the probability that no guest will receive the proper hat.

Solution. 0.375

Probability



Exercise 25

Suppose each day that you drive to work a traffic light that you encounter is either green with probability $7/16$, red with probability $7/16$, or yellow with probability $1/8$, independent of the status of the light on any other day. If over the course of five days, G , Y , and R denote the number of times the light is found to be green, yellow, or red, respectively,

- (a) What is the probability that $P(G = 2, Y = 1, R = 2)$?
- (b) Also, what is the probability $P(G = R)$?

Solution. (a) 0.1374 (b) 0.1449

Exercise 26

Is it possible for A and B to be independent events yet satisfy $A = B$?

Solution. $A = B = S$ or $A = B = \emptyset$

Probability



Exercise 27

In an experiment, A , B , C , and D are events with probabilities $P(A) = 1/4$, $P(B) = 1/8$, $P(C) = 5/8$, and $P(D) = 3/8$. Furthermore, A and B are disjoint, while C and D are independent.

(a) Find $P(A \cap B)$, $P(A \cup B)$, $P(A \cap \overline{B})$, and $P(A \cup \overline{B})$.

(b) Are A and B independent?

(c) Find $P(C \cap D)$, $P(C \cap \overline{D})$, and $P(\overline{C} \cap \overline{D})$.

(d) Are $\overline{C}$ and $\overline{D}$ independent?

Solution. (a) 0 $3/8$ $1/4$ $7/8$ (b) Yes (c) $15/64$ $25/64$ $15/64$ (d) Yes

Probability



Exercise 28

Suppose that traffic engineers have coordinated the timing of two traffic lights to encourage a run of green lights. In particular, the timing was designed so that with probability 0.75 a driver will find the second light to have the same color as the first. Assuming the first light is equally likely to be red or green.

- (a) What is the probability that the second light is green?
- (b) What is the probability that you wait for at least one light?

Solution. (a) 0.5 (b) 0.625

Probability



Exercise 29

Consider a binary code with 5 bits (0 or 1) in each code word. An example of a code word is 01010. In each code word, a bit is a zero with probability 0.8, independent of any other bit.

- (a) What is the probability of the code word 00111?
- (b) What is the probability that a code word contains exactly three ones?

Solution. (a) 0.00512 (b) 0.0512

Probability



Exercise 30

A rat maze consists of a straight corridor, at the end of which is a branch; at the branching point the rat must either turn right or left. Assume 10 rats are placed in the maze, one at a time.

- (a) If each is choosing one of the two branches at random, what is the distribution of the number that turn right?
- (b) What is the probability at least 9 will turn the same way?

Exercise 31

An airline sells 200 tickets for a certain flight on an airplane that has only 198 seats because, on the average, 1 percent of purchasers of airline tickets do not appear for the departure of their flight. Determine the probability that everyone who appears for the departure of this flight will have a seat.

Solution. 0.5954

Probability



Exercise 32

A student who is trying to write a paper for a course has a choice of two topics, A and B . If topic A is chosen, the student will order 2 books through interlibrary loan, while if topic B is chosen, the student will order 4 books. The student feels that a good paper necessitates receiving and using at least half the books ordered for either topic chosen.

- (a) If the probability that a book ordered through interlibrary loan actually arrives on time is 0.85 and books arrive independently of one another, which topic should the student choose to maximize the probability of writing a good paper?
- (b) What if, the arrival probability is only 0.55 instead of 0.85?

Probability



Exercise 33

Two different suppliers, A and B, provide a manufacturer with the same part. All suppliers of this part are kept in a large bin. In the past, 5 percent of the parts supplied by A and 9 percent of the parts supplied by B have been defective. A supplies four times as many parts as B. Suppose you reach into the bin and select a part and find it is non-defective. What is the probability that it was supplied by A?

Solution. 0.8067

Exercise 34

An urn contains 2 black and 5 brown balls. A ball is selected at random. If the ball drawn is brown, it is replaced and 2 additional brown balls are also put into the urn. If the ball drawn is black, it is not replaced in the urn and no additional balls are added. A ball is then drawn from the urn the second time. What is the probability that the ball selected at the second stage is brown?

Solution. 0.793651

Probability



Exercise 35

A factory has three machines A, B, and C. Past records show that the machine A produced 40% of the items of output, the machine B produced 35% of the items of output, and machine B produced 25% of the items. Further 2% of the items produced by machine A were defective, 1.5% produced by machine B were defective, and 1% produced by machine B were defective.

- (a) If an item is drawn at random, what is the probability that it is defective?
- (b) An item is acceptable if it is not defective. What is the probability that an acceptable item comes from machine A?